Jieke Shi

Research Scientist | NRF Postdoctoral Award Recipient
Singapore Management University

Address: College of Integrative Studies, B2,
RISE Center, 10 Canning Rise, Singapore 179873

E-Mail: jiekeshi@smu.edu.sg

Website: https://jiekeshi.tech

RESEARCH INTERESTS

Software Engineering, Program Synthesis, Performance Engineering, Software Testing, LLMs

I design and implement **automated synthesis**-assisted methods for **performance optimization** of code model serving and **safety validation** of software systems.

43 published or accepted papers, including 15 as first, co-first, or corresponding author (October 2026). Research contributions include **360+ bugs and vulnerabilities detected and fixed** in projects including GCC, Clang, OpenSSL, curl, FFmpeg, and OP-TEE.

EDUCATION

Aug 2022 – Mar 2026 **Singapore Management University**, Ph.D. in Computer Science
Supervisor: David Lo
Dissertation: *Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis*
Defense: 16 March 2026

Aug 2016 – Jun 2020 **Yangzhou University**, B.Eng. in Electronic Engineering

EMPLOYMENT

Mar 2026 – Present **Singapore Management University**, Research Scientist
NRF Postdoctoral Award recipient; Principal Investigator of an independent research project supported by a S$250,000 research grant.

Jun 2021 – Mar 2026 **Singapore Management University**, Research Engineer
Lead research staff under the AI.SG Programme on testing reinforcement-learning-enabled systems.

Jul 2020 – Apr 2021 **Institute of Computing Technology, Chinese Academy of Sciences**, Research Assistant Intern

PUBLICATIONS

Note: In software engineering, ICSE, ASE, and FSE are recognized as top-tier conferences, and TOSEM and TSE are top-tier journals (CORE A*); ISSTA, MSR, ICSME, and SANER are prominent conferences covering software testing, mining, analysis, and evolution (CORE A). In security, S&P is a top-tier conference (CORE A*), while ACSAC is a leading venue with a focus on system security applications (CORE A). In natural language processing, ACL is a top-tier conference (CORE A*).

[C]: Full Conference Paper (Total: **28**)
[S]: Short Paper (Total: **6**)
[J]: Full Journal Paper (Total: **9**)
[P]: Preprint (Total: **2**)

Preprints

[P.2] **Jieke Shi**, Junda He, Wenjia Jiang, Weifeng Sun, Shidong Pan, Zhensu Sun, Chengran Yang, Peixin Zhang, Yifan Jia, Zhou Yang, Thong Hoang, Xiwei Xu, Zhenchang Xing, and David Lo. *"Lossless Tensor Compression as Program Synthesis"*. arXiv preprint arXiv:2608.02162, 2026, 9 pages. [PDF] [Code]

[P.1] Bo Wang, Pengyang Wang, Chong Chen, Ming Deng, **Jieke Shi**, Qi Sun, Chengran Yang, Youfang Lin, Zhou Yang, Junjie Chen, Jun Sun, and David Lo. *"Mut4All: Fuzzing Compilers via LLM-Synthesized Mutators Learned from Bug Reports"*. arXiv preprint arXiv:2507.19275, 2025, 28 pages. [PDF] [Code]

2027

[C.28] Chengyan Ma, **Jieke Shi**, Ruidong Han, Weifeng Sun, Yuqing Niu, Yunbo Lyu, Zhou Yang, and David Lo. *"Missing Security-Fix Backporting for GitHub Actions Workflows"*. In Proceedings of the 49th IEEE/ACM International Conference on Software Engineering (**ICSE 2027**).

[C.27] Junda He, Xiangyu Shan, **Jieke Shi**, Yunbo Lyu, Gou Tan, Xiuwei Shang, Zhou Yang, Christoph Treude, and David Lo. *"When Agreement Hides Disagreement: Are We Evaluating LLM Judges Fairly?"*. In Proceedings of the 49th IEEE/ACM International Conference on Software Engineering (**ICSE 2027**).

2026

[C.26] Huihui Huang, **Jieke Shi (Corr.)**, Bo Wang, Zhou Yang, and David Lo. *"Finding Memory Leaks in C/C++ Programs via Neuro-Symbolic Augmented Static Analysis"*. In Proceedings of the 43rd Annual Computer Security Applications Conference (**ACSAC 2026**, Technical Paper), 16 pages. [PDF] [Code]

[J.9] Chengyan Ma, Ruidong Han, **Jieke Shi (Corr.)**, Ye Liu, Yuqing Niu, Di Lu, Chuang Tian, Jianfeng Ma, Debin Gao, and David Lo. *"DITING: A Static Analyzer for Identifying Bad Partitioning Issues in TEE Applications"*. IEEE Transactions on Software Engineering (**TSE 2026**). [PDF] [Code]

[C.25] Gou Tan, Zhensu Sun, **Jieke Shi**, Ting Zhang, Zilong He, Qingfu Wu, Shuai Liang, Weifeng Sun, Junda He, Pengfei Chen, Chuanfu Zhang, Lwin Khin Shar, and David Lo. *"AgentChaos: Chaos Engineering for Agent Systems via Programmatic Fault Injection"*. In Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**, Research Track).

[C.23] Weifeng Sun, **Jieke Shi**, Zhou Yang, Yuchen Chen, Hongyan Li, Meng Yan, and David Lo. *"How Reasoning Shapes Social Bias in LLM-Generated Code?"*. In Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**, Research Track).

[C.22] Xiuwei Shang, Li Hu, Xiao Jiang, **Jieke Shi (Corr.)**, Junda He, Zhou Yang, Shaoyin Cheng, Guoqiang Chen, Weiming Zhang, and David Lo. *"Beyond Text Matching: Towards Reference-Free Evaluation for Human-Oriented Binary Reverse Engineering"*. In Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering (**ASE 2026**, Research Track).

[C.21] Thong Hoang, Mykhailo Klymenko, Xiwei Xu, Shidong Pan, Yi Ding, Xushuo Tang, Zhengyi Yang, **Jieke Shi**, and David Lo. *"ESG Reporting Lifecycle Management with Large Language Models and AI Agents"*. In Proceedings of the 23rd IEEE International Conference on Software Architecture (**ICSA 2026**, Software Architecture in Practice). [PDF]

[C.20] **Jieke Shi**, Junda He, Zhou Yang, Chengran Yang, Mykhailo Klymenko, Thong Hoang, Sherry Xu, Zhenchang Xing, and David Lo. *"Compiling Code LLMs into Lightweight Executables"*. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 24 pages. [PDF] [Code] **Featured by SMU Research News**

[C.19] Chengyan Ma, **Jieke Shi (Corr.)**, Ruidong Han, Ye Liu, Feng Li, Yuqing Niu, and David Lo. *"Automated Repair of TEE Partitioning Issues via DSL-Guided and LLM-Assisted Patching"*. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 24 pages.

[C.18] Yuqing Niu, **Jieke Shi (✉ Corr.)**, Ruidong Han, Ye Liu, Chengyan Ma, Yunbo Lyu, and David Lo. *"What You Trust Is Insecure: Demystifying How Developers (Mis)Use Trusted Execution Environments in Practice"*. In Proceedings of the 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2026, 🏆IEEE Computer Society TCSE Distinguished Paper Award**). [PDF] [Code]

[C.17] Bo Wang, Yueyang Chen, **Jieke Shi**, Minghui Li, Yunbo Lyu, Yinan Wu, Youfang Lin, and Zhou Yang. *"Hidden Licensing Risks in the LLMware Ecosystem"*. In Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2026**, Research Track). [PDF] [Code]

[C.24] Junda He, **Jieke Shi**, Zhou Yang, Mingfei Cheng, and David Lo. *"Learning from the Test: Self-Referential Differential Testing for Deep RL Agents"*. In Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2026**, Research Track).

[S.6] Chengyan Ma, **Jieke Shi**, Ruidong Han, Ye Liu, Yuqing Niu, and David Lo. *"Finding Missing Input Validation in TEEs via LLM-Assisted Symbolic Execution"*. In Proceedings of the 3rd ACM International Conference on AI Foundation Models and Software Engineering (**FORGE 2026**, New Idea Papers).

[C.16] Chengran Yang, Zhensu Sun, Hong Jin Kang, **Jieke Shi**, and David Lo. *"Think Like Human Developers: Harnessing Community Knowledge for Structured Code Reasoning"*. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026, 🏆ACM SIGSOFT Distinguished Paper Award**). [PDF]

[C.15] Huihui Huang, Ratnadira Widyasari, Ting Zhang, Ivana Clairine IRSAN, **Jieke Shi (✉ Corr.)**, Han Wei Ang, Frank Liauw, Eng Lieh Ouh, Lwin Khin Shar, Hong Jin Kang, and David Lo. *"Back to the Basics: Rethinking Issue-Commit Linking with LLM-Assisted Retrieval"*. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**). [PDF] [Code]

[J.8] Junda He, **Jieke Shi**, Terry Yue Zhuo, Christoph Treude, Jiamou Sun, Zhenchang Xing, Xiaoning Du, and David Lo. *"LLM-as-a-Judge for Software Engineering: Literature Review, Vision, and the Road Ahead"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**). [PDF]

[S.5] Huihui Huang, **Jieke Shi**, Junkai Chen, Ting Zhang, Yikun Li, Chengran Yang, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. *"PenForge: On-the-Fly Expert Agent Construction for Automated Penetration Testing"*. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, NIER Track). [PDF] [Code]

[C.14] Junkai Chen, Huihui Huang, Yunbo Lyu, Junwen An, **Jieke Shi**, Chengran Yang, Ting Zhang, Haoye Tian, Yikun Li, Zhenhao Li, Xin Zhou, Xing Hu, and David Lo. *"SecureVibeBench: Evaluating Secure Coding Capabilities of Code Agents with Realistic Vulnerability Scenarios"*. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**, Main Conference). [PDF]

[C.13] Chengran Yang, Ting Zhang, Jinfeng Jiang, Xin Zhou, Haoye Tian, Mingzhe Du, **Jieke Shi**, Junkai Chen, Yikun Li, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. *"SeCuRepair: Semantics-Aligned, Curriculum-Driven, and Reasoning-Enhanced Vulnerability Repair Framework"*. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026**, Main Conference, Oral, 🏆**SAC Highlight Award**). [PDF]

2025

[J.7] **Jieke Shi**, Junda He, Zhou Yang, Đorđe Žikelić, and David Lo. *"Synthesizing Efficient and Permissive Programmatic Runtime Shields for Neural Policies"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [PDF] [Code]

[J.6] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, Dongsun Kim, DongGyun Han, and David Lo. *"Finding Safety Violations of AI-Enabled Control Systems through the Lens of Synthesized Proxy Programs"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [PDF] [Code]

[J.5] **Jieke Shi**, Zhou Yang, and David Lo. *"Efficient and Green Large Language Models for Software Engineering: Literature Review, Vision, and the Road Ahead"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [PDF]

[J.4] Zhou Yang, **Jieke Shi (📧 Corr.)**, Premkumar Devanbu, and David Lo. *"Ecosystem of Large Language Models for Code"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [PDF] [Code]

[C.12] Yunbo Lyu, Zhou Yang, **Jieke Shi**, Jianming Chang, Yue Liu, and David Lo. *""My productivity is boosted, but ..." Demystifying Users' Perception on AI Coding Assistants"*. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2025, 🏆ACM SIGSOFT Distinguished Paper Award**). [PDF] [Code]

[J.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, Bowen Xu, Xin Zhou, DongGyun Han, and David Lo. *"Prioritizing Speech Test Cases"*. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [PDF] [Code]

2024

[J.2] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. *"Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models"*. IEEE Transactions on Software Engineering (**TSE 2024**). [PDF]

[C.11] **Jieke Shi**, Zhou Yang, Hong Jin Kang, Bowen Xu, Junda He, and David Lo. *"Greening Large Language Models of Code"*. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]

[J.1] Zhou Yang, Bowen Xu, Jie M. Zhang, Hong Jin Kang, **Jieke Shi**, Junda He, and David Lo. *"Stealthy Backdoor Attack for Code Models"*. IEEE Transactions on Software Engineering (**TSE 2024, 🏆 IEEE Computer Society Best Paper Award**). [PDF] [Code]

[C.10] Chen Gong, Zhou Yang, Yunpeng Bai, Junda He, **Jieke Shi**, Kecen Li, Arunesh Sinha, Bowen Xu, Xinwen Hou, David Lo, and Tianhao Wang. *"BAFFLE: Hiding Backdoors in Offline Reinforcement Learning Datasets"*. In Proceedings of the 45th IEEE Symposium on Security and Privacy (**S&P 2024**). [PDF] [Code]

[C.9] Junda He, Zhou Yang, **Jieke Shi**, Chengran Yang, Kisub Kim, Bowen Xu, Xin Zhou, and David Lo. *"Curiosity-Driven Testing for Sequential Decision-Making Process"*. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]

[C.8] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. *"Unveiling Memorization in Code Models"*. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]

2023

[C.7] Zhou Yang, Chenyu Wang, **Jieke Shi**, Thong Hoang, Pavneet Kochhar, Qinghua Lu, Zhenchang Xing, and David Lo. *"What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues"*. In Proceedings of the 20th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2023**). [PDF] [Code]

2022

[C.6] **Jieke Shi**, Zhou Yang, Bowen Xu, Hong Jin Kang, and David Lo. *"Compressing Pre-trained Models of Code into 3 MB"*. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022, 🎖ACM SIGSOFT Distinguished Paper Award Nomination**). [PDF] [Poster] [Code]

[S.4] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, and David Lo. *"Can Identifier Splitting Improve Open-Vocabulary Language Model of Code?"*. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**, ERA Track). [PDF] [Poster] [Code]

[C.5] Chen Gong, Zhou Yang, Yunpeng Bai, **Jieke Shi**, Arunesh Sinha, Bowen Xu, David Lo, Xinwen Hou, Guoliang Fan. *"Curiosity-Driven and Victim-Aware Adversarial Policies"*. In Proceedings of the 39th Annual Computer Security Applications Conference (**ACSAC 2022,** 🏆 **Honorable Mention Award**). [PDF] [Code]

[C.4] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. *"Revisiting Neuron Coverage Metrics and Quality of Deep Neural Networks"*. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [PDF] [Code]

[C.3] Zhou Yang, **Jieke Shi**, Junda He, and David Lo. *"Natural Attack for Pre-trained Models of Code"*. In Proceedings of the 44th IEEE/ACM International Conference on Software Engineering (**ICSE 2022**). [PDF] [Code]

[C.2] Chengran Yang, Bowen Xu, Ferdian Thung, Yucen Shi, Ting Zhang, Zhou Yang, Xin Zhou, **Jieke Shi**, Junda He, DongGyun Han, and David Lo. *"Answer Summarization for Technical Queries: Benchmark and New Approach"*. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**). [PDF] [Code]

[C.1] Ratnadira Widyasari, Stefanus Agus Haryono, Ferdian Thung, **Jieke Shi**, Constance Tan, Fiona Wee, Jack Phan, and David Lo. *"On the Influence of Biases in Bug Localization: Evaluation and Benchmark"*. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [PDF] [Code] [Dataset]

2021

[S.3] Zhou Yang ⚖️, **Jieke Shi** ⚖️, Shaowei Wang, and David Lo. *"IncBL: Incremental Bug Localization"*. In Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2021**, Tool Demonstrations). [PDF] [Poster] [Code]

[S.2] Zhou Yang, Harshit Jain, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. *"BiasHeal: On-the-Fly Black-Box Healing of Bias in Sentiment Analysis Systems"*. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2021**, NIER Track). [PDF] [Code]

[S.1] Muhammad Hilmi Asyrofi, Zhou Yang, **Jieke Shi**, Chu Wei Quan, and David Lo. *"Can Differential Testing Improve Automatic Speech Recognition Systems?"*. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2021**, NIER Track). [PDF] [Code]

HONORS & AWARDS

2026 **National Research Foundation (NRF) Postdoctoral Award** (S$250,000), Singapore NRF

ACM SIGSOFT Distinguished Paper Award, ICSE 2026

IEEE Computer Society TCSE Distinguished Paper Award, SANER 2026

SAC Highlight Award, ACL 2026

Distinguished Reviewer Award, AIware 2026

Distinguished PC Member Award, ICSME 2026 (Visions and Emerging Results Track)

2025 **ACM SIGSOFT Distinguished Paper Award**, ASE 2025

IEEE Computer Society Best Paper Award, IEEE Transactions on Software Engineering

OpenAI Researcher Access Program Grant ($5,000)

Google Cloud Research Credits Grant ($2,500)

Research Staff Excellence Awards (awarded to only 4 researchers university-wide), Singapore Management University

SMU SCIS PhD Research Excellence Award, Singapore Management University
SMU SCIS Dean's List Award, Singapore Management University

2023 **Presidential Doctoral Fellowship**, Singapore Management University
SCIS Dean's List Award, Singapore Management University

2022 **Honorable Mention Award**, ACSAC 2022
ACM SIGSOFT Distinguished Paper Award Nomination, ASE 2022

2020 **Outstanding Undergraduate Thesis** (Top 0.2% in province), Jiangsu Province, China
Outstanding Graduate (Top 5%), Yangzhou University

2019 & Before **National Encouragement Scholarship** (Top 3%), Ministry of Education of China
2nd Class Presidential Scholarship, Yangzhou University
1st Class Presidential Scholarship, Yangzhou University

ACADEMIC SERVICES

Program Committee (PC)/Reviewer

2027 **AAAI Conference on Artificial Intelligence** (AAAI), PC
2027 **IEEE International Conference on Software Analysis, Evolution and Reengineering** (SANER), PC
2027 **Mining Software Repositories Conference** (MSR), PC
2027 **IEEE/ACM International Conference on Program Comprehension** (ICPC), PC in Research Track
2027 **Agentic Engineering Workshop** (AGENT@ICSE), PC
2026 **International Conference on the Foundations of Software Engineering** (FSE), PC in Ideas, Visions and Reflections Track
2026 **Workshop on Human-Centered AI for Software Engineering** (HumanAISE), PC
2026 **ICML Workshop on Deep Learning for Code** (DL4C@ICML), PC
2026 **International Conference on AI-powered Software** (AIware), PC in Main Track & Benchmark and Dataset Track
2026 **International Conference on Software Maintenance and Evolution** (ICSME), PC in Research Papers & Visions and Emerging Results Track
2025–2027 **International Conference on Software Engineering** (ICSE), Shadow PC
2024–2026 **Mining Software Repositories Conference** (MSR), Junior PC
2025–2026 **International Workshop on Green and Sustainable Software** (GREENS@ICSE), PC
2026 **European Conference on Computer Systems** (EuroSys), Shadow PC & Artifact Evaluation
2026 **International Conference on Evaluation and Assessment in Software Engineering** (EASE), PC in AI Models/Data Track
2025 **Annual Meeting of the Association for Computational Linguistics** (ACL), Reviewer
2025 **IEEE International Computer Software and Applications Conference** (COMPSAC), PC

Organizing Committee

2024 **International Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems** (SEA4DQ@FSE), Web Chair

Journal Reviewer

2025–Present **IEEE Transactions on Dependable and Secure Computing** (TDSC)
2025–Present **Empirical Software Engineering** (EMSE)
2023–Present **ACM Transactions on Software Engineering and Methodology** (TOSEM)

2023–Present **IEEE Transactions on Software Engineering** (TSE)
2021–Present **Neurocomputing** (Software Section)

External Reviewer

2021–2025 **International Conference on Software Engineering** (ICSE)
2022–2026 **IEEE/ACM International Conference on Automated Software Engineering** (ASE)
2023–2025 **ACM International Conference on the Foundations of Software Engineering** (ESEC/FSE)
2023 **International Conference on Software Maintenance and Evolution** (ICSME)
2023 **International Workshop on LLMs for Code** (LLM4Code@ICSE)
2022–2023 **International Joint Conference on Artificial Intelligence** (IJCAI)
2022 **IEEE International Conference on Software Analysis, Evolution and Reengineering** (SANER)
2022 **International Conference on AI Engineering – Software Engineering for AI** (CAIN)
2022 **International Workshop on Software Engineering for Responsible AI** (SE4RAI@ICSE)
2022 **International Conference on Engineering of Complex Computer Systems** (ICECCS)

INVITED TALKS

Dec 2025 **Shanghai Jiao Tong University**, Research Seminar
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis

Nov 2025 **Singapore Programming Languages Summit 2025**, Invited Talk
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis

Nov 2025 **A-Mobile Workshop of ASE 2025**, Keynote, Invited by Guozhu Meng
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis

Nov 2025 **Korea University**, Workshop on the Next Generation of CodeLLMs and Their Applications, Invited by Dongsun Kim
Optimizing and Fortifying AI Software via Program Synthesis

Sep 2025 **Singapore Management University**, RISE-WASP Workshop, Invited by Lingxiao Jiang
Fortifying Autonomous Systems via Program Synthesis

Nov 2023 **Singapore Management University**, Research Cluster Seminar, Invited by David Lo
Towards Compact and Efficient Language Models of Code via Compression

Oct 2023 **North Carolina State University**, Guest Lecture, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [Link]

TEACHING EXPERIENCE

Jan 2026 **Singapore Management University**, Teaching Assistant in CS706, Invited by David Lo
Software Mining and Analysis, Graduate Core Course

Oct 2023 **North Carolina State University**, Guest Lecture in CSC791, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [Link]

Jan 2020 - Jun 2020 **Yangzhou University**, Teaching Assistant
Digital Signal Processing, Undergraduate Core Course

GRANT PROPOSAL WRITING EXPERIENCE

2025 - 2027 **ESG-based Responsible AI: Toward Green, Secure, and Compliant LLM Utilisation for Digital Service Development Process**
CSIRO - A*STAR Partnership Program, Singapore & Australia
Total Award: Over SGD $500K

2024 - 2028 **TrustedSEERs: Trusted Intelligent Work Bots for Engineering Better Software**
National Research Foundation (NRF), Singapore
Total Award: Over SGD $3.2M

2023 - 2025 **TitanCA: Vulnerability Detection using Large Code Models**
Government Technology Agency (GovTech), Singapore
Total Award: Over SGD $3.8M

REFEREES

David Lo, ACM/IEEE/ASE Fellow, ACM Distinguished Speaker, OUB Chair Professor
School of Computing and Information Systems, Singapore Management University
davidlo@smu.edu.sg

Zhou Yang, Canada CIFAR AI Chair Nominee, Amii Fellow, Assistant Professor
Department of Computer Science, University of Alberta
zy25@ualberta.ca